

PRATIK MANDANI

PRODUCT DEVELOPMENT & EMBEDDED QUALITY ENGINEER

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PROFESSIONAL SUMMARY

Results-driven **Product Development & Embedded Quality Engineer** with 7+ years of hands-on experience in industrial product testing, hardware prototyping, PCB validation, and rigorous QA/QC workflows. Proven track record in component-level troubleshooting, precision SMD rework, and cross-functional integration with PLC, Raspberry Pi, and embedded firmware teams. Combines deep hardware validation background with 4+ years of full-stack web software engineering (Laravel, MySQL, REST APIs) to build reliable, end-to-end industrial automation systems.

CORE TECHNICAL SKILLS

Hardware & Quality Engineering: Electronic Instruments Testing, QA/QC Protocols, Component-Level Troubleshooting, Incoming Inspection, Calibration & Metrology, PCB Validation, Precision Hand Soldering & SMD Rework.

Embedded & IoT Systems: C, C++, Arduino IDE, ESP32, Raspberry Pi Integration, Sensor Interfacing, Microcontroller Programming, QR-Code Automation.

Web & Software Development: PHP, Laravel, MySQL Architecture, RESTful APIs, HTML5, CSS3, JavaScript, jQuery, Bootstrap.

Tools & Methodologies: Git (Version Control), Agile/Scrum Methodologies, System Diagnostics, Production Line Optimization.

PROFESSIONAL EXPERIENCE

Logic & Web Systems Engineer (Laravel Developer) | Altius Infoway — Rajkot, India May 2022 – Present

- Architected, developed, and maintained scalable backend logic and MySQL database structures for 100+ live web platforms.
- Engineered custom ERP/CRM administrative platforms, streamlining data pipelines and optimizing queries for high performance.
- Constructed responsive front-end user interfaces utilizing HTML5, CSS3, JavaScript, and Bootstrap following modern UX standards.
- Integrated third-party APIs and resolved legacy software communication bottlenecks within Agile development lifecycles.

Product Development & R&D Engineer | Symbian Vending Solutions — Rajkot, India Nov 2018 – Mar 2022

- Led electro-mechanical assembly, functional quality testing, and field deployment for industrial automated vending machines.
- Executed component-level hardware diagnostics, electrical circuit analysis, and comprehensive QA/QC validation protocols.
- Soldered and reworked prototype PCBs, replaced precision SMD components, and repaired field-return units with rapid turnaround.
- Designed and programmed Arduino and ESP32 microcontroller systems for QR-code-driven automated dispensing systems.
- Collaborated closely with PLC, Raspberry Pi, and backend software engineering teams to ensure seamless hardware-software integration.
- Successfully delivered and validated 200+ production units within a strict 10-day manufacturing schedule through optimized workflow.

KEY PROJECTS & PORTFOLIOS

- Smart Dispensing & Industrial Automation Systems:** Prototyped and validated automated vending controllers using ESP32/Arduino microcontrollers, sensor arrays, and QR-code dispensing mechanisms.
- Enterprise & Industrial Web Platforms:** Developed dynamic database architectures and management software for industrial brands including Diamorivo Systems, Pisa Hardware, Akash Electric Company, and Lakme Vitrified.

EDUCATION

Diploma in Electronics & Communication Engineering | Gujarat Technological University (GTU) Passed Out: 2018

ADDITIONAL DETAILS

Languages: English, Hindi, Gujarati

Core Strengths: QA/QC, Troubleshooting, Cross-functional Leadership

Availability / Notice Period: Immediate / 30 Days

Focus Areas: Embedded Systems, IoT, Hardware & Web Integration